Tech Dessert #12: Feed Lines and Connectors

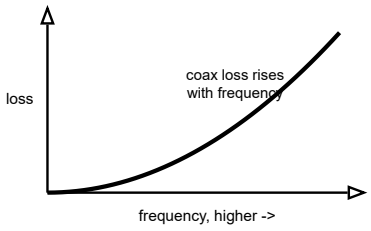
Companion to Bite #12 -- the coaxial cable and connectors that carry your signal from the radio to the antenna.

Group T9B · 12 pool questions · Self-study handout · Practice exams at eham.net/exams

This Dessert is the companion to Bite 12, and the last sheet in the program. Bite 12 covered the antenna; this one covers the **feed line** – the cable between the radio and that antenna. Read the table, study the diagram, then drill the question bank.

Feed Lines and Connectors

Topic	What to know	Q
Most common coax impedance in amateur radio	**50 ohms.**	T9B02
Why coax is the most common feed line	It is **easy to use** and needs few special installation considerations.	T9B03
What “SWR” measures	How well a load (the antenna) is **matched** to the transmission line.	T9B12
What an antenna tuner (coupler) does	**Matches** the antenna system’s impedance to the transceiver’s output.	T9B04
What happens to coax loss as frequency rises	The loss **increases**.	T9B05
The lowest-loss feed line	**Air-insulated hardline.**	T9B11
RG-58 versus RG-213 coax	**RG-213 has less loss** at a given frequency (it is the thicker cable).	T9B10
Sources of loss in a coax feed line	Water in the connectors, high SWR, and multiple connectors – **all of these**.	T9B08
Best connector for above 400 MHz	**Type N.**	T9B06
What is true of PL-259 connectors	They are **commonly used at HF and VHF** frequencies.	T9B07
Connectors that need taping for weather	PL-259, BNC, and Type N – **all of these**, when used outdoors.	T9B01
What causes erratic changes in SWR	A **loose connection** in the antenna or feed line.	T9B09



Coaxial cable loses more signal as frequency rises – so on VHF and UHF, short runs of good, low-loss coax matter.

Remember: amateur coax is **50 ohms**, and it is popular because it is **easy to install**. Its loss **rises with frequency**, so on VHF and UHF use good cable (RG-213 beats RG-58; hardline is best) and keep runs short. **SWR** measures the antenna-to-line match; an **antenna tuner** improves the match the radio sees. Keep **water out of connectors**, and remember that erratic SWR usually means a **loose connection**. That is the final Dessert – the whole exam is now behind you.

Dessert 12 Question Bank (T9B)

T9B01. Which of the following connectors should be carefully taped for weather protection when used outdoors?
A. PL259
B. BNC
C. Type N
D. All these choices are correct

T9B02. What is the most common impedance of coaxial cables used in amateur radio?
A. 8 ohms
B. 50 ohms
C. 600 ohms
D. 12 ohms

T9B03. Why is coaxial cable the most common feed line for amateur radio antenna systems?
A. It is easy to use and requires few special installation considerations
B. It has less loss than any other type of feed line
C. It can handle more power than any other type of feed line
D. It is less expensive than any other type of feed line

T9B04. What is the major function of an antenna tuner (antenna coupler)?
A. It matches the antenna system impedance to the transceiver's output impedance
B. It helps a receiver automatically tune in weak stations
C. It allows an antenna to be used on both transmit and receive
D. It automatically selects the proper antenna for the frequency band being used

T9B05. What happens as the frequency of a signal in coaxial cable is increased?
A. The characteristic impedance decreases
B. The loss decreases
C. The characteristic impedance increases
D. The loss increases

T9B06. Which of the following connector types is most suitable as an RF connector for frequencies above 400 MHz?
A. PL-259
B. Type N
C. RS-213
D. DB-25

T9B07. Which of the following is true of PL-259 type coax connectors?
A. They are preferred for microwave operation
B. They are watertight
C. They are commonly used at HF and VHF frequencies
D. They are a bayonet-type connector

T9B08. Which of the following is a source of loss in coaxial feed line?
A. Water intrusion into coaxial connectors
B. High SWR
C. Multiple connectors in the line
D. All these choices are correct

T9B09. What can cause erratic changes in SWR?
A. Local thunderstorm
B. Loose connection in the antenna or feed line
C. Over-modulation
D. Overload from a strong local station

T9B10. What is the electrical difference between RG-58 and RG-213 coaxial cable?
A. There is no significant difference between the two types
B. RG-58 cable has two shields
C. RG-213 cable has less loss at a given frequency
D. RG-58 cable can handle higher power levels

T9B11. Which of the following types of feed line has the lowest loss?
A. 50-ohm flexible coax
B. Multi-conductor unbalanced cable
C. Air-insulated hardline
D. 75-ohm flexible coax

T9B12. What is standing wave ratio (SWR)?
A. A measure of how well a load is matched to a transmission line
B. The ratio of amplifier power output to input
C. The transmitter efficiency ratio
D. An indication of the quality of your station's ground connection

Answer Key

T9B01: D	T9B02: B	T9B03: A	T9B04: A
T9B05: D	T9B06: B	T9B07: C	T9B08: D
T9B09: B	T9B10: C	T9B11: C	T9B12: A